

Part C

Course title: SC7 Bayes Methods - OPEN BOOK
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version**

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1. (a) [7 marks] Let $\theta \in \Omega$ be a parameter with prior density $\pi(\theta)$. Let $y = (y_1, \dots, y_n)$, $y \in R^n$ be a realisation of data Y with observation model $Y \sim p(\cdot|\theta)$.
- (i) Explain what it means for $\pi(\theta|y)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $Y = y$.

Solution: (2 marks - bookwork) $\pi(\theta|y) \propto p(y|\theta)\pi(\theta)$. This is DI if either $p(y|\theta)$ or $\pi(\theta)$ are intractable functions of θ . This may happen if for example the normalisation of $\pi(y, \theta)$ is not available in closed form. MCMC targeting $\pi(\theta|y)$ does not require the overall normalisation $p(y)$ for $\pi(y|\theta)\pi(\theta)$ as it cancels in the Metropolis ratio $\pi(y|\theta')\pi(\theta')/\pi(y|\theta)\pi(\theta)$. However if $\pi(y|\theta)$ or $\pi(\theta)$ are intractable then this cannot be evaluated and so MH-MCMC is not possible.

- (ii) Let $S(y) = (S_1(y), \dots, S_p(y))$ be a p -component vector of statistics $S : R^n \rightarrow R^p$ computed from the data. For $S, S' \in R^p$ let $d(S, S')$ be a distance function and let $\delta > 0$ be a simulation parameter. Let $\Delta_y(\delta) = \{y' \in R^n : d(S(Y), S(y)) < \delta\}$. Write down a rejection-ABC algorithm simulating $\pi(\theta|Y \in \Delta_y(\delta))$.

Solution: (2 marks - bookwork)

Step 1. Simulate $\theta \sim \pi(\theta)$ and $y' \sim p(y'|\theta)$.

Step 2. If $d(S(y), S(y')) < \delta$ then stop and return θ else goto step 1.

- (iii) Prove that your rejection-ABC algorithm simulates $\pi(\theta|Y \in \Delta_y(\delta))$.

Solution: (3 marks - discussed briefly in lectures) The algorithm simulates $(\theta, y') \sim \pi(\theta)p(y'|\theta)$ conditioned on $y' \in \Delta_y(\delta)$, so the joint distribution is

$$\theta, y'|y' \in \Delta_y(\delta) \sim \frac{\pi(\theta)p(y'|\theta)}{\int \mathbb{I}_{z \in \Delta_y(\delta)} \pi(\theta)p(z|\theta) dz d\theta}.$$

We obtain the marginal distribution for θ by integrating dy' over the set $\Delta_y(\delta)$ to get

$$\theta, |y' \in \Delta_y(\delta) \sim \frac{\pi(\theta)p(y' \in \Delta_y(\delta)|\theta)}{p(y' \in \Delta_y(\delta))}$$

which is equal $\pi(\theta|y' \in \Delta_y(\delta))$ as supposed.

- (b) [9 marks] The spread of a virus is modeled as a contact process on a square lattice with cells $\mathcal{C} = \{(i, j) : i, j \in \mathbb{Z}\}$. Cells are activated (ie, infected) by their neighbors. Let $\mathcal{N}_{(i,j)} = \{(i-1, j), (i+1, j), (i, j-1), (i, j+1)\}$ be the neighbors of cell (i, j) . Let $0 \leq r \leq 1$ be the recovery probability and $0 \leq a \leq 1/4$ be the activation probability. The set of active cells $X^t \subset \mathcal{C}$ at times $t \in (0, 1, \dots, T)$ is simulated as follows.

Simulation of $X \sim \text{contact}(a, r)$: Let $X_0 = \{(0, 0)\}$. For t from 0 to $T-1$

Update Set $X^{t+1} = X^t$. Let $N_{(i,j)}^t = |\mathcal{N}_{(i,j)} \cap X^t|$ give the number of active neighbors of cell (i, j) at time t . Let $E^t = \{(i, j) \in \mathcal{C} \setminus X^t : N_{(i,j)}^t > 0\}$ be the list of exposed cells.

Activate For $(i, j) \in E^t$ set $X^{t+1} \leftarrow X^{t+1} \cup \{(i, j)\}$ with probability $aN_{(i,j)}^t$.

Deactivate For $(i, j) \in X^t$ set $X^{t+1} \leftarrow X^{t+1} \setminus \{(i, j)\}$ with probability r .

Let $X = X^T$ be the set of all cells active at time T .

- (i) Suppose we observe $X_{obs} \sim \text{contact}(a, r)$ with a, r unknown but given prior $\pi(a, r)$. Show that $\pi(a, r|X_{obs})$ is in general doubly intractable.

Solution: (3 marks - new) The posterior is

$$\pi(r, a|X_{obs}) \propto p(X_{obs}|r, a)\pi(a, r).$$

For $A \subset \mathcal{C}$ let ∂A be the cells outside A adjacent a cell in A (so $\partial A = \bigcup_{(i,j) \in A} \mathcal{N}_{(i,j)} \setminus A$). Then $(X_0, X_1, \dots, X_T)$ is a possible sequence of active sites if it only adds active sites at the boundary so $X_t \setminus X_{t-1} \subseteq \partial X_{t-1}$ for $t = 1, \dots, T$, and we must start at $X_0 = \{(0, 0)\}$ and end at $X_T = X_{obs}$. Let χ be the set of all possible sequences, given the data. Then

$$p(X_{obs}|r, a)\pi(a, r) = \sum_{(X_0, X_1, \dots, X_T) \in \chi} \prod_{t=1}^T P(X_t|X_{t-1}).$$

Now $P(X_t|X_{t-1})$ is easily computed in terms of a, r given the two sets X_{t-1} and X_t . However the sum over $(X_0, X_1, \dots, X_T)$ will be intractable if T is at all large (even if X_{obs} has few elements) as cells can deactivate and reactivate.

- (ii) Give a rejection-ABC algorithm targeting $\pi(r, a|X \in \Delta_{X_{obs}}(\delta))$. Your answer should define and justify summary-statistics and distance measure.

Solution: (6 marks - standard (the ABC) and new (the summary statistic)) For $(a, r) \sim \pi(a, r)$ and $X \sim \text{contact}(a, r)$, let summary statistics $S(X) \in R^p$ be given. I will assume these are scaled a posteriori so that $\text{var}(S_i(X)) \simeq 1$ so they vary on similar scales. Take the distance $d(S, S') = \|S - S'\|$. Fix $\delta > 0$ (chosen by experiment so that a small non zero fraction of samples below are accepted). The Rejection-ABC algorithm is

For $j = 1, \dots, J$

Step 1. Simulate $(a, r) \sim \pi(a, r)$ and $X \sim \text{contact}(a, r)$

Step 2. If $d(S(X_{obs}), S(X)) \leq \delta$ stop and return $(a^{(j)}, r^{(j)}) = (a, r)$.

Otherwise, goto Step 1.

This generates a set of samples $(a^{(j)}, r^{(j)})$, $j = 1, \dots, J$ distributed according to $\pi(r, a|X \in \Delta_{X_{obs}}(\delta))$ with $\Delta_{X_{obs}}(\delta) = \{X \in \chi : d(S(X_{obs}), S(X)) \leq \delta\}$.

How to choose S ? Clearly the number of elements $|X_{obs}|$ is sensitive to both a and r . The length of the boundary $|\partial X_{obs}|$ will tend to be large if r is at all large, as “interior” cells will appear. The number of disjoint sets in X and the number of “holes” in X might be informative. A reasonable answer might take $S(X) = (|X|, |\partial X|)$ as a starting point for experiment.

- (c) [9 marks] If $\pi(\theta)$ is a prior for θ then an inference scheme is a rule $\psi(\theta; \pi, y)$ for updating belief for θ given data y . For example in Bayesian inference $\psi_{\text{Bayes}}(\theta; \pi, y) = \pi(\theta|y)$ but in ABC at fixed δ , $\psi_{\Delta, \delta}(\theta; \pi, y) = \pi(\theta|Y \in \Delta_y(\delta))$.

For $1 \leq j < n$ let $y_{1:j} = (y_1, \dots, y_j)$ and $y_{j+1:n} = (y_{j+1}, \dots, y_n)$. Suppose the data are conditionally independent, so

$$p(y|\theta) = \prod_{i=1}^n p(y_i|\theta).$$

A belief update is *order-coherent* for conditionally independent data if

$$\psi(\theta; \pi, y) = \psi(\theta; \psi(\theta; \pi, y_{1:j}), y_{j+1:n})$$

for all $j \in \{1, 2, \dots, n-1\}$ (the posterior from the first data set is the prior for the next).

(i) Show that Bayesian inference is order-coherent.

Solution: (2 marks - new) In this case $\psi(\theta; \pi, y_{1:j}) \propto p(y_{1:j}|\theta)\pi(\theta)$ so

$$\psi(\theta; \psi(\theta; \pi, y_{1:j}), y_{j+1:n}) \propto p(y_{j+1:n}|\theta)p(y_{1:j}|\theta)\pi(\theta)$$

which is $\pi(\theta|y)$ by the conditional independence.

(ii) Show that ABC with fixed δ is not in general order coherent. *Hint: take summary statistic $S(y) = y$ and Euclidean distance measure $d(y, y') = \|y - y'\|$ and give a counter-example.*

Solution: (4 marks - new) In this case $\psi(\theta; \pi, y_{1:j}) \propto p(Y_{1:j} \in \Delta_{y_{1:j}}(\delta)|\theta)\pi(\theta)$ so

$$\psi(\theta; \psi(\theta; \pi, y_{1:j}), y_{j+1:n}) \propto p(Y_{j+1:n} \in \Delta_{y_{j+1:n}}(\delta)|\theta)p(Y_{1:j} \in \Delta_{y_{1:j}}(\delta)|\theta)\pi(\theta)\pi(\theta).$$

This is order coherent iff

$$p(Y_{j+1:n} \in \Delta_{y_{j+1:n}}(\delta)|\theta)p(Y_{1:j} \in \Delta_{y_{1:j}}(\delta)|\theta) \propto p(Y \in \Delta_y(\delta)|\theta)$$

For the given distance

$$p(Y \in \Delta_y(\delta)|\theta) = \Pr(A + B < \delta^2|\theta)$$

with $A = |Y_1 - y_1|^2 + \dots + |Y_j - y_j|^2$ and $B = |Y_{j+1} - y_{j+1}|^2 + \dots + |Y_n - y_n|^2$ so the condition is

$$\Pr(A + B < \delta^2|\theta) = c \Pr(A < \delta^2|\theta) \Pr(B < \delta^2|\theta).$$

for some $c > 0$. This cant hold in general. For example if $Y_i \sim \text{Bernoulli}(\theta)$ for $i = 1, 2$ so $n = 2$ and $y_1 = y_2 = 0$ then $A + B \sim \text{Binomial}(2, \theta)$. If $\delta^2 = 1.5$ then

$$\Pr(A + B < \delta^2|\theta) = \Pr(Y_1 + Y_2 < 1.5|\theta) = (1 - \theta)^2 + 2\theta(1 - \theta)$$

but $\Pr(A < \delta^2|\theta) = \Pr(Y_1 < 1.5|\theta) = \Pr(B < \delta^2|\theta) = 1$.

(iii) Let $C_y(\delta)$ be the rectangular prism $C_y(\delta) = \{y' \in R^n : |y_i - y'_i| < \delta \forall i = 1, \dots, n\}$. Show that inference with $\psi_{C,\delta}(\theta; \pi, y) = \pi(\theta|Y \in C_y(\delta))$ is order-coherent.

Solution: (3 marks - new) In this case $\psi(\theta; \pi, y_{1:j}) \propto p(Y_{1:j} \in B_{y_{1:j}}(\delta)|\theta)\pi(\theta)$ so

$$\psi(\theta; \psi(\theta; \pi, y_{1:j}), y_{j+1:n}) \propto p(Y_{j+1:n} \in B_{y_{j+1:n}}(\delta)|\theta)p(Y_{1:j} \in B_{y_{1:j}}(\delta)|\theta)\pi(\theta)\pi(\theta).$$

This is order coherent since

$$p(Y_{j+1:n} \in B_{y_{j+1:n}}(\delta)|\theta)p(Y_{1:j} \in B_{y_{1:j}}(\delta)|\theta) = p(Y \in B_y(\delta)|\theta)$$

because $Y \in B_y(\delta)$ iff $Y_{1:j} \in B_{y_{1:j}}(\delta)$ and $(Y_{j+1:n} \in B_{y_{j+1:n}}(\delta))$, and $Y_{1:j}$ and $Y_{j+1:n}$ are independent.

2. Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables. For $n = 1, 2, 3, \dots$ the probability mass functions are $\Pr\{X_1 = x_1, \dots, X_n = x_n\} = p(x_1, \dots, x_n)$.

(a) [10 marks]

- (i) What does it mean to say that these variables are exchangeable?

Solution: (1 mark - bookwork) The sequence is exchangeable if

$$p(x_1, x_2, \dots, x_n) = p(x_{\sigma_1}, x_{\sigma_2}, \dots, x_{\sigma_n})$$

for each $n \geq 1$ and any permutation $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ of the numbers $(1, 2, \dots, n)$.

- (ii) State de Finetti's theorem for the sequence $X_1, X_2, X_3, \dots$

Solution: (2 marks - bookwork) There exists a distribution function $F(\theta)$ on $[0, 1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 \left[\prod_{i=1}^n p(x_i | \theta) \right] dF(\theta)$$

where $p(x_i | \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}$. The random variable

$$\Theta = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N X_i \quad \text{exists and} \quad F(\theta) = \Pr(\Theta \leq \theta).$$

- (iii) Let S be a sample space and let $\mathcal{S}$ be a σ -field of sets in S . In the following events A, B etc are subsets of S and members of $\mathcal{S}$.

For events A, B write $A < B$ if A is less likely a priori than B . Let $\sim$ indicate equality in preference.

Write down the first three of the Savage axioms, in the form given by de Groot.

Solution: (3 marks bookwork) Axiom 1. For any two events A and B exactly one of the following relations must hold: $A > B$, $A < B$, $A \sim B$.

Axiom 2. If $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \geq B_i, i = 1, 2$ then $A_1 \cup A_2 \geq B_1 \cup B_2$.

If in addition either $A_1 > B_1$ or $A_2 > B_2$ then $A_1 \cup A_2 > B_1 \cup B_2$.

Axiom 3. If $A \in \mathcal{S}$ then $\emptyset \leq A$.

- (iv) Suppose $A \subseteq B$ (as sets). Show from the axioms that $A \leq B$ (in preference). You may quote results from lectures and classes without proof.

Solution: (4 marks - S) We need a lemma: if $A_1 \cap D = B_1 \cap D = \emptyset$ then $A_1 \cup D > B_1 \cup D$ iff $A_1 > B_1$. This was in a problem sheet (though the inequality wasn't strict there) so can be quoted without proof. To show this take $A_2 = B_2 = D$ in axiom 2. Then $A_1 > B_1$ implies $A_1 \cup D > B_1 \cup D$ (last part of A2). On the other hand if $A_1 \leq B_1$ and $A_2 = B_2 = D$ then $A_1 \cup D \leq B_1 \cup D$ (first part of A2). Meaning: we can add the same set to both sides of an inequality, if it doesn't intersect the other sets, and we can remove the same set from both sides.

Now suppose $A > B$. By the lemma we can remove A from both sides to get $\emptyset > B \setminus A$ which is a contradiction by Axiom 3.

(b) [15 marks]

(i) Consider the sequence of probability mass functions given for $n = 1, 2, 3, \dots$ by

$$p_n(x) = c_n 2^{-k_n(n-k_n)},$$

with $x = (x_1, \dots, x_n)$, $k_n = \sum_{i=1}^n x_i$ and c_n a normalising constant. Suppose that for each $n = 1, 2, 3, \dots$ our prior probability $P(A)$ for the outcome $A = \{(x_1, \dots, x_n)\}$ is $p_n(x_1, \dots, x_n)$. Prove that the associated prior preferences are not coherent.

Hint: $c_3 = 2/7, c_4 = 8/27$.

Solution: (6 marks - new but similar to last year's question) These preferences do not satisfy the Savage axioms. Take $A_1 = 0000, A_2 = 0001$ and $B_1 = 000$ and $B_2 = \emptyset$. Since $p(000) = 2/7, p(0000) = 8/27$ and $8/27 > 2/7$ we have the preference $0000 > 000$ so $A_1 > B_1$. By the third axiom $A_2 \geq B_2$. Now $A_1 \cap A_2 = \emptyset = B_1 \cap B_2$ so $A_1 \cup A_2 > B_1 \cup B_2$ by the second axiom. But $A_1 \cup A_2 = 000$ and $B_1 \cup B_2 = 000$ so we have $000 > 000$, a contradiction.

(ii) Show that there does not exist an infinite exchangeable sequence of binary random variables $Y_1, Y_2, Y_3, \dots$ satisfying $\Pr\{Y_1 = x_1, \dots, Y_n = x_n\} = p_n(x_1, \dots, x_n)$ for $n \geq 1$.

Solution: (3 marks - new) EITHER - We know that there doesn't exist a joint distribution with the stated marginals (as the Savage axioms are not satisfied). If there was an IES with these marginals then a joint distribution would be given by de Finetti, a contradiction, so no such IES exists.
OR - Alternatively could show that the stated distributions are not marginally consistent. If an IES $Y_1, Y_2, Y_3, \dots$ exists with PMF's $\Pr\{Y_1 = x_1, \dots, Y_n = x_n\} = p(x_1, \dots, x_n)$ then these PMF's must be marginally consistent. For example, we must have $p(000) = p(0000) + p(0001)$. However, $p_3(000) = 2/7$ but $p_4(0000) + p_4(0001) = 8/27(1 + 1/4) = 10/27$. These are not equal, so the given PMF's are not marginally consistent, and hence there can't exist rv's $Y_1, \dots, Y_4$ with the stated marginals. Same argument could be used to show preferences are incoherent.

(iii) Consider the sequence of probability mass functions given for $n = 1, 2, 3, \dots$ by

$$q_n(x) = \frac{k_n!(n-k_n)!}{(n+1)!},$$

with $x = (x_1, \dots, x_n)$ and $k_n = \sum_{i=1}^n x_i$ as before. Suppose that for each $n = 1, 2, 3, \dots$ our prior probability $Q(A)$ for the event $A = \{(x_1, \dots, x_n)\}$ is $q_n(x_1, \dots, x_n)$. Prove that the associated prior preferences are coherent.

Hint: is there an infinite exchangeable sequence with these marginals?

Solution: (6 marks - new) It is not practical to verify the Savage axioms (we only wrote down the first three). However, if we can give an IES with these marginals, then the joint distribution of rv in the sequence assigns coherent preferences so the Savage axioms must be satisfied.
If $q \sim U(0, 1)$ and $X_i \sim \text{Bernoulli}(q)$ iid for $i = 1, \dots, n$ then $X_1, X_2, X_3, \dots$ is an infinite exchangeable sequence by construction, or if we prefer, this follows by

de Finetti since the marginal probabilities, $p(x_1, \dots, x_n)$ say, are given by

$$p(x_1, \dots, x_n) = \int_0^1 \prod_{i=1}^n q^{x_i} (1-q)^{1-x_i} U(q; 0, 1) dq.$$

However this sequence is our sequence since

$$p(x_1, \dots, x_n | q) = p^k (1-p)^{n-k}$$

where $k = \sum_i x_i$ and

$$\begin{aligned} p(x_1, \dots, x_n) &= \int_0^1 q^{k_n} (1-q)^{n-k_n} dq \\ &= \frac{k_n! (n-k_n)!}{(n+1)!}. \end{aligned}$$

3. (a) [5 marks] A random probability distribution $G \sim DP(\alpha, H)$ has a Dirichlet Process distribution with parameter $\alpha > 0$ and base distribution H on $\Omega = \mathbb{R}^p$.
- (i) Define the Dirichlet Process $DP(\alpha, H)$.

Solution: (2 marks - bookwork) $G \sim DP(\alpha, H)$ iff $\forall$ measurable partitions $A_1, A_2, \dots, A_r$ of Ω

$$G(A_1), \dots, G(A_r) \sim \text{Dirichlet}(\alpha H(A_1), \dots, \alpha H(A_r))$$

- (ii) Calculate the expectation and variance of $G(A)$, $A \subseteq \Omega$ in terms of α and $H(A)$, and briefly interpret the role α and H play in determining the distribution of G .

$$\text{If } X \sim \text{Beta}(a, b) \text{ then } E(X) = \frac{a}{a+b} \text{ and } \text{var}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Solution: (3 marks - bookwork) First, $G(A), G(A^c) \sim \text{Beta}(a, b)$ with $a = \alpha H(A)$ and $b = \alpha H(A^c)$ by the agglomerative property of Dirichlet distributions. We have

$$E(G(A)) = \frac{a}{a+b} = \frac{\alpha H(A)}{\alpha H(A) + \alpha H(A^c)} = H(A)$$

since $H(A) + H(A^c) = 1$. The variance is

$$\begin{aligned} \text{var}(G(A)) &= \frac{ab}{(a+b)^2(a+b+1)} \\ &= \frac{\alpha^2 H(A)H(A^c)}{(\alpha H(A) + \alpha H(A^c))^2(\alpha H(A) + \alpha H(A^c) + 1)} \\ &= \frac{H(A)(1-H(A))}{\alpha + 1} \end{aligned}$$

Here H is the “centre” of the distribution of G in the sense that it is the mean about which G varies, and α controls the variance of G around H . When α is large, G is similar in distribution to H .

- (b) [10 marks] Consider a linear mixed model with a Dirichlet-Normal mixture model for the random effects. For $i = 1, \dots, n$ let $x_i, y_i \in \mathbb{R}$ be covariate and response. For $i = 1, \dots, n$ the observation model is

$$y_i = \alpha_i + x_i \beta_i + \epsilon_i, \quad \text{with } \epsilon_i \sim N(0, \sigma^2)$$

independently. Let $y = (y_1, \dots, y_n)$. Let $\theta_i = (\alpha_i, \beta_i)$ for $i = 1, \dots, n$ and $\theta = (\theta_1, \dots, \theta_n)$. The θ -prior is $\theta_i \sim G$ independent for $i = 1, \dots, n$ with $G \sim DP(\alpha, H)$. The distribution $H(\cdot; \psi)$ has density $h(\cdot; \psi)$, parameters ψ and $\alpha > 0$ is fixed. A prior $\pi(\sigma, \psi)$ is given. Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ list the distinct values in θ with $\theta_k^* = (\alpha_k^*, \beta_k^*)$, $k = 1, \dots, K$. Let $S = (S_1, \dots, S_K)$ be a partition of $\{1, 2, \dots, n\}$ satisfying $i \in S_k$ if $\theta_i = \theta_k^*$ and let $n_k = |S_k|$. Let $P_\alpha(S)$ be the prior probability for partition S .

- (i) Interpret the generative model for the data.

Solution: (2 marks - new) Individual observations come from K different groups. Within each group $k = 1, \dots, K$ there is a simple normal linear relation between y and x with $E(y) = \alpha + \beta x$. However this linear relation varies between groups by both slope and intercept. This is essentially a mixture of regressions. The observation noise is the same for all groups. The number of groups, and the assignment S of observations to groups, is unknown.

- (ii) Suggest an explicit form for the density h (depending on a parameter vector ψ which you should give). Your choice is illustrative, there is no need to justify it.

Solution: (2 marks - standard) The density h of the base distribution H should have support on $\Omega = \mathbb{R}^2$. A typical choice would be

$$h(\theta_i; \psi) = N(\alpha_i; \mu_\alpha, \sigma_\alpha^2) N(\beta_i; \mu_\beta, \sigma_\beta^2)$$

with $\psi = (\mu_\alpha, \sigma_\alpha, \mu_\beta, \sigma_\beta)$ for any $\theta_i = (\alpha_i, \beta_i) \in \mathbb{R}^2$.

- (iii) Write down the posterior $\pi(\sigma, \psi, \theta^*, S|y)$ in as much detail as you can.

$$\text{You may use without proof that } P_\alpha(S) = \frac{\alpha^K \prod_{k=1}^K (n_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)}.$$

Solution: (5 marks - standard)

$$\pi(\sigma, \psi, \theta^*, S|y) \propto \pi(\sigma, \psi) P_\alpha(S) \pi(\theta^*|S, \psi) L(\theta^*, \sigma; y)$$

Here $\pi(\sigma, \psi)$ is given, $P_\alpha(S)$ is given in the hint and

$$\pi(\theta^*|S, \psi) = \prod_{k=1}^K h(\theta_k^*; \psi),$$

with $h(\theta_k^*; \psi) = N(\alpha_i^*; \mu_\alpha, \sigma_\alpha^2) N(\beta_i^*; \mu_\beta, \sigma_\beta^2)$ for my choice of H above and more generally $h(\theta_k^*; \psi)$ when $H(d\theta_k^*; \psi) = h(\theta_k^*; \psi) d\theta_k^*$. The likelihood is

$$L(\theta^*, \sigma; y) = \prod_{k=1}^K \prod_{i \in S_k} N(y_i; \alpha_k^* + x_i \beta_k^*, \sigma^2).$$

Let Ξ_n be the set of all partitions of $\{1, 2, \dots, n\}$ and $S \in \Xi_n$ let $K(S) = K$ when $S = \{S_1, \dots, S_K\}$. The parameter space, $(\sigma, \psi, \theta^*, S) \in \Omega^*$ say, is

$$\Omega^* = [0, \infty) \times \Omega_\psi \times \left[\bigcup_{S \in \Xi_n} \mathbb{R}^{2K(S)} \times \{S\} \right],$$

with $\psi \in \Omega_\psi$ the parameter space of ψ , in my case $\Omega_\psi = \mathbb{R} \times [0, \infty) \times \mathbb{R} \times [0, \infty)$.

- (c) [10 marks] Consider a reversible jump MCMC algorithm targeting $\pi(\sigma, \psi, \theta^*, S|y)$.

- (i) Let $P(S'|S)$ give the probability to propose S' given S . Choose $i \sim U\{1, 2, \dots, n\}$. Let $S^{-i} = (S_1^{-i}, \dots, S_{K-i}^{-i})$ be the partition obtained from $S = (S_1, \dots, S_K)$ by removing element i from its set $i \in S_{k_i}$ in S . Here $K_{-i} = K - 1$ if we create an empty

cluster when we remove i and otherwise $K_{-i} = K$. Choose $k \in \{1, 2, \dots, K^{-i} + 1\}$ with probability p_k , $k = 1, \dots, K^{-i} + 1$, where

$$(p_1, \dots, p_{K^{-i}}, p_{K^{-i}+1}) \propto (|S_1^{-i}|, \dots, |S_{K^{-i}}^{-i}|, \alpha).$$

Assign i to set S_k^{-i} : if $k \leq K^{-i}$ set $S' = (S_1^{-i}, \dots, S_k^{-i} \cup \{i\}, \dots, S_{K^{-i}}^{-i})$; if $k = K^{-i} + 1$ set $S' = (S_1^{-i}, \dots, S_{K^{-i}+1}^{-i})$ with $S_{K^{-i}+1}^{-i} = \{i\}$. Suppose $S' \neq S$. Show that

$$P(S'|S) = n^{-1} \frac{P_\alpha(S')}{P_\alpha(S^{-i})}.$$

Solution: (5 marks - standard - minor variant on PS4 problem) From the process, $P(S'|S) = n^{-1} P(S'|S^{-i}, i)$ since this is the probability to choose i and then create S' from S^{-i} . We are given $P(S'|S^{-i}, i) = p_k$ where k is the cluster in S' containing i , since we had to add i to that cluster in S^{-i} .

Then EITHER ... We recognise p_k is the vector of table assignment probabilities for the last arrival in the CRP. This process is exchangeable so we may make i the last arrival. It follows that $P_\alpha(S'|S^{-i}) = p_k$ is the conditional probability for the final seating in the CRP given the penultimate seating. But $S'^{-i} = S^{-i}$ since the two partitions agree except for i , and so $p_k = P_\alpha(S')/P_\alpha(S^{-i})$.

OR ... work back from the answer and verify that the given p_k satisfy $p_k = P_\alpha(S')/P_\alpha(S^{-i})$ by direct substitution into P_α given in the hint.

(ii) Let $X_t = (\sigma, \psi, \theta^*, S)$. Simulate $S' \sim P(S'|S)$ as above.

- If $K' = K$ set $\tilde{\theta}^* = \theta^*$.

- If $K' = K - 1$ set $\tilde{\theta}^* = \theta_{-k_i}^*$, the vector with the parameter for cluster $k_i = \{k : i \in S_k\}$ removed.

- If $K' = K + 1$ set $\tilde{\theta}^* = (\theta^*, \dots, \theta_K^*, \theta_{K+1}^*)$ with $\theta_{K+1}^* \sim h(\cdot; \psi)$

With probability $\alpha(\tilde{\theta}^*, S'|\theta^*, S)$ set $X_{t+1} = (\sigma, \psi, \tilde{\theta}^*, S')$ else $X_{t+1} = (\sigma, \psi, \theta^*, S)$.

Calculate $\alpha(\tilde{\theta}^*, S'|\theta^*, S)$ for the case $K' = K + 1$.

Solution: (5 marks - new) Quoting the general form for the acceptance probability in reversible jump

$$\alpha(\tilde{\theta}^*, S'|\theta^*, S) = \min \left\{ 1, \frac{\pi(\sigma, \psi, \tilde{\theta}^*, S'|y) q(\tilde{\theta}^*, S'|\tilde{\theta}^*, S')}{\pi(\sigma, \psi, \theta^*, S|y) q(\tilde{\theta}^*, S'|\theta^*, S)} \right\}.$$

Here $\tilde{\theta}_k^* = (\alpha_k^*, \beta_k^*)$, $k = 1, \dots, K$ and $\tilde{\theta}_{K+1}^* = (\alpha_{K+1}^*, \beta_{K+1}^*)$ with $(\alpha_{K+1}^*, \beta_{K+1}^*) \sim h$. In the acceptance probability above,

$$\pi(\sigma, \psi, \tilde{\theta}^*, S'|y) \propto \pi(\sigma, \psi) P_\alpha(S') \prod_{k=1}^{K+1} \left[h(\tilde{\theta}_k^*|\psi) \prod_{i \in S_k} N(y_i; \alpha_k^* + x_i \beta_k^*, \sigma^2) \right],$$

$$q(\tilde{\theta}^*, S'|\theta^*, S) = n^{-1} \frac{P_\alpha(S')}{P_\alpha(S^{-i})} h(\theta_{K+1}^*|\psi),$$

$$\pi(\sigma, \psi, \theta^*, S|y) \propto \pi(\sigma, \psi) P_\alpha(S) \prod_{k=1}^K \left[h(\theta_k^*|\psi) \prod_{i \in S_k} N(y_i; \alpha_k^* + x_i \beta_k^*, \sigma^2) \right],$$

and

$$q(\theta^*, S|\theta^*, S') = n^{-1} \frac{P_\alpha(S)}{P_\alpha(S'^{-i})},$$

and note $S'^{-i} = S^{-i}$ in the last relation. Substituting these into the expression for $\alpha(\tilde{\theta}^*, S'|\theta^*, S)$ and canceling,

$$\alpha(\tilde{\theta}^*, S'|\theta^*, S) = \min \left\{ 1, \frac{N(y_i; \alpha_{K+1}^* + x_i \beta_{K+1}^*, \sigma^2)}{N(y_i; \alpha_{k_i}^* + x_i \beta_{k_i}^*, \sigma^2)} \right\}.$$

It is straightforward to check (and not needed for the answer) that the Jacobian is one and dimensions balance.